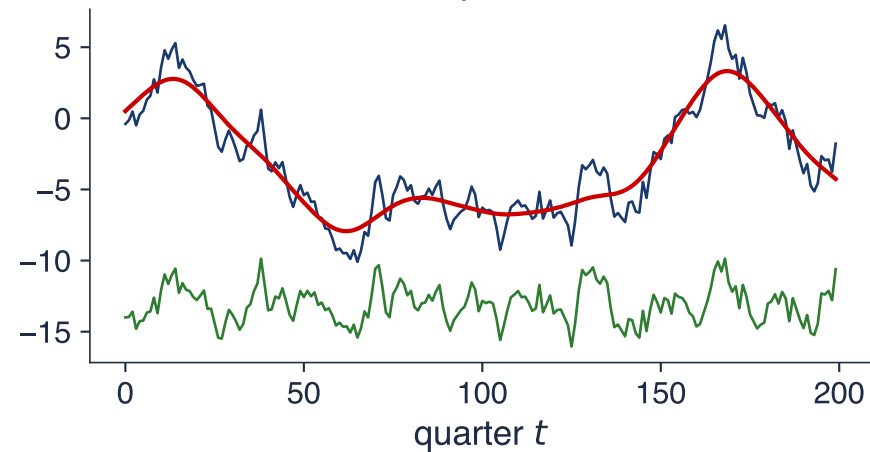
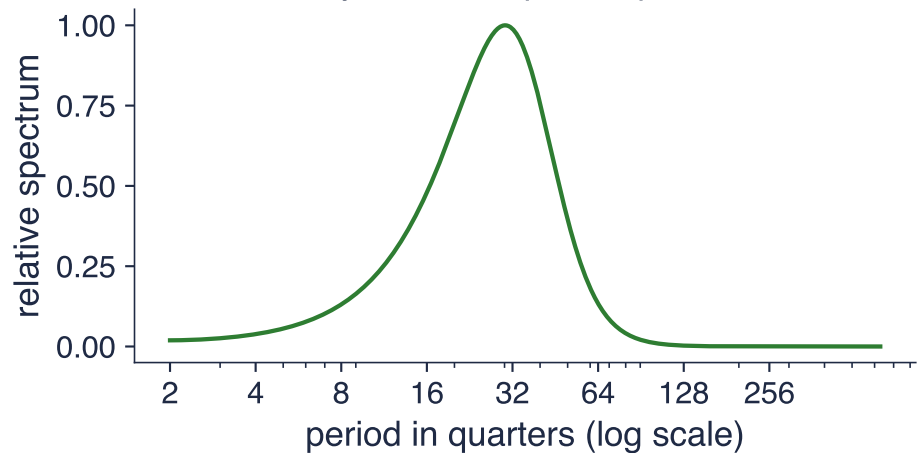


A random walk has no cycle...



...but its HP cycle has a spectral peak



— random walk y_t — HP trend $\hat{\tau}_t$ — HP cycle $y_t - \hat{\tau}_t$ (shifted)